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Project Report
on

“Solar Power Bank Charger For Mobile”

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Battery Management System-(23EEE532)

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Abstract

The rapid growth of smartphones and portable electronic devices has significantly increased global energy consumption, creating a strong demand for reliable, portable, and sustainable charging solutions. In many parts of the world, particularly in rural and remote regions, access to continuous electrical power is still limited. Even in urban areas, frequent travel, outdoor activities, and emergency situations make conventional charging methods unreliable. In this context, solar-powered charging systems have emerged as a promising and environmentally friendly alternative. This project, Solar Panel Power Bank for Mobile Charging, aims to design, develop, and analyze a compact, efficient, and cost-effective power bank that utilizes solar energy as its primary source of power generation.The proposed system integrates three major components: a photovoltaic (PV) solar panel, a rechargeable lithium-ion battery pack, and an advanced Battery Management System (BMS). The PV panel captures solar radiation and converts it into direct current (DC) electricity through the photovoltaic effect. This energy is then conditioned using an appropriate charge controller or DC–DC converter to ensure that the energy supplied to the battery is stable and within the optimal charging range. The inclusion of a charge controller prevents over-voltage and under-voltage stresses on the battery, thereby improving charging efficiency and extending the overall lifespan of the storage unit.The energy generated from the solar panel is stored in a high-capacity lithium-ion battery, which serves as an energy reservoir for mobile charging. The battery is configured with a BMS that plays a critical role in ensuring efficient and safe operation. The BMS monitors key parameters such as cell voltage, charging current, temperature, and state of charge (SOC). It protects the battery from overcharging, over-discharging, short circuits, overcurrent conditions, and overheating. By balancing the voltage across the cells, the BMS enhances battery reliability, ensures long-term stability, and prevents hazardous conditions such as thermal runaway.Once the energy is stored, the power bank delivers regulated output voltages, typically 5V through USB ports, making it compatible with standard mobile devices. The system is designed to provide a stable output even during fluctuating solar input conditions. This ensures uninterrupted charging for smartphones and other gadgets. Additionally, the device can be charged both through solar energy and traditional power supply (dual charging mode), increasing convenience and usability during low-light or non-sunny days.A significant advantage of this solar power bank is its portability, which allows users to carry it anywhere, making it an ideal solution for students, travelers, trekkers, campers, and individuals living in off-grid areas.

The design prioritizes low cost and practical implementation so that the system can be adopted widely. The use of renewable energy makes the project environmentally sustainable, reducing carbon emissions and decreasing dependence on fossil fuel-based electricity generation. Furthermore, by integrating a compact solar panel with a robust energy storage system, the project demonstrates how small-scale renewable energy solutions can contribute to global clean energy goals.Overall, this project showcases a practical, eco-friendly, and efficient approach to mobile charging. It addresses real-world challenges related to energy accessibility and portable power availability. The proposed solar-powered power bank highlights the potential of renewable energy technologies in day-to-day applications and encourages the adoption of green energy solutions at the consumer level. Through careful design, testing, and performance evaluation, the project illustrates how solar energy can be transformed into a reliable and user-friendly power source for modern mobile communication devices.

INTRODUCTION

In today's modern world, portable electronic devices such as smartphones, tablets, and wearables have become an essential part of daily life. These devices require constant power for communication, entertainment, and work-related activities. However, frequent charging through conventional grid electricity can be inconvenient, especially in remote or outdoor environments where power outlets are not readily available. To address this challenge, the concept of a solar-powered power bank offers an efficient, eco-friendly, and self-sustaining solution for charging mobile devices anywhere under sunlight.

A solar power bank is a compact energy storage device that integrates a solar panel, rechargeable lithium battery, and power management circuit to convert, store, and deliver electrical energy for mobile charging. The system harnesses solar energy, converts it into electrical energy through the photovoltaic (PV) effect, and stores it in rechargeable lithium-ion cells. When required, this stored energy is boosted and regulated to supply a stable 5 V DC output compatible with USB-powered devices.

In this project, a 134N3P C-Type USB Power Bank Module is used as the core control and conversion unit. This module integrates several important functions such as battery charging, discharging, voltage boosting, overcharge and short-circuit protection, and provides both Type-C input/output and standard USB-A output ports. It efficiently converts the 3.7 V nominal voltage of a single lithium-ion battery to the required 5 V for mobile charging. The module's built-in battery management system (BMS) ensures safe charging and discharging operations, preventing overvoltage, undervoltage, and current surges that could damage the cells or connected devices.

The project uses a 3.7 V 18650 lithium-ion rechargeable battery with a capacity of 3000 mAh as the main energy storage element. This cell type is known for its high energy density, lightweight design, and long cycle life, making it ideal for portable energy storage systems. The solar panel connected to the module supplies the necessary input power to charge the battery during daylight hours. The size and power rating of the solar panel are selected to provide sufficient charging current for the battery while maintaining portability.

By integrating these components, the system forms a renewable and sustainable mobile charging solution that reduces dependency on the conventional electricity grid and promotes the use of clean solar energy. The proposed design is low-cost, compact, and suitable for outdoor applications such as camping, travel, or emergency situations.

Thus, the Solar Power Bank Charger using 134N3P Module, Solar Panel, and 18650 Battery demonstrates the practical application of solar energy in consumer electronics. It combines renewable energy harvesting, energy storage, and efficient DC-DC conversion into one user-friendly portable device, contributing to both energy conservation and technological convenience.

LITERATURE REVIEW

Several studies and prototype projects have been carried out on solar-powered power banks that integrate solar panels, rechargeable lithium batteries, and USB output modules. These works highlight the importance of renewable energy utilization for portable electronic devices and demonstrate that solar energy can be effectively used to charge mobile phones, especially in remote or outdoor areas.

Researchers have shown that 18650 lithium-ion batteries are ideal for such applications due to their high energy density, long cycle life, and low self-discharge rate. Many studies emphasize that safe charging and discharging require a Battery Management System (BMS) to prevent overcharge, over-discharge, and short-circuit conditions.

Earlier designs often used TP4056 charging modules with boost converters for 5V USB output. However, these were found to be inefficient when directly powered by a solar panel due to fluctuating input voltage. To overcome this, new modules like the 134N3P C-Type USB power bank module integrate both charging, boosting, and protection functions in a single circuit, providing stable 5V output and supporting both USB-A and Type-C connections.

Studies also indicate that the performance of a solar power bank depends largely on the solar panel rating, sunlight availability, and module efficiency. Small panels (3–5 W) can charge single 18650 cells but take longer times, while higher-rated panels (10–20 W) achieve faster recharging and better overall efficiency.

In summary, literature on solar power banks suggests that combining a 3.7V 18650 (3000 mAh) lithium-ion battery, a 134N3P power management module, and a solar panel provides a compact, efficient, and eco-friendly charging solution for mobile devices. This combination ensures energy storage, voltage regulation, and device protection while utilizing clean solar ener

COMPONENTS DETAILS

1. 134N3P C-Type USB Power Bank Module

The 134N3P C-Type USB Power Bank Module **is** a compact and efficient circuit board used for charging and discharging single 3.7V lithium-ion batteries such as 18650 cells. It integrates a charging controller, boost converter, and protection circuit in one unit, making it ideal for solar power bank applications. The module takes a 5V input through a Type-C or Micro USB port—this can come from a USB charger or a regulated solar panel—and safely charges the battery up to 4.2V

During use, the module boosts the 3.7V battery voltage to a stable 5V output through its built-in DC-DC converter, providing up to 2A current for powering or charging mobile devices. It also includes a Battery Management System (BMS) that prevents overcharge, over-discharge, overcurrent, and short-circuit damage, ensuring battery safety and long life. Four LED indicators show the battery's charge level, and the system automatically shuts off when the battery is fully charged or deeply discharged.

In short, the 134N3P module serves as a complete power management solution for portable solar power banks, combining charging, protection, and 5V output functions efficiently in one small board.

2. Lithium Rechargable Battery

The 3.7V 18650 3000mAh lithium-ion rechargeable battery is one of the most widely used power storage cells in portable and renewable energy applications due to its high energy density, efficiency, and long lifespan. The name "18650" denotes its physical dimensions—18 mm in diameter and 65 mm in length—which makes it compact yet capable of storing a significant amount of energy. It delivers a nominal voltage of 3.7 volts and can be charged up to a maximum of 4.2 volts, typically using a constant-current/constant-voltage (CC/CV) charging method. With a capacity of 3000mAh (or 3 ampere-hours), it can provide about 11.1 watt-hours of energy, which is sufficient to power small electronic gadgets, LED lights, and mobile devices when integrated into a power bank.

These batteries are lightweight and have a low self-discharge rate, meaning they retain charge for extended periods when not in use. They also offer high discharge currents, allowing them to support devices that require steady power output. In a solar power bank charger system, the 3.7V 18650 battery stores the electrical energy generated by the solar panel, managed through a charging module such as the 134N3P USB power bank module. This module regulates the input from the solar panel, ensuring safe charging and providing stable 5V USB output for mobile charging.

For safety, most 18650 batteries include protection circuits to prevent overcharging, over-discharging, overheating, or short circuits—conditions that can otherwise damage the battery or reduce its lifespan. They can typically withstand 500–1000 charge-discharge cycles before their capacity starts to degrade. Because of their reliability and efficiency, these batteries are commonly found not only in power banks but also in laptops, flashlights, electric vehicles, and solar-powered systems. Overall, the 3.7V 18650 3000mAh lithium-ion battery is a robust and versatile energy storage option, perfectly suited for use in solar-based mobile charging applications.

3. Solar Panel

A 5V mini solar panel is a compact photovoltaic module designed to generate around 5 volts of output, making it suitable for charging small electronic devices and powering modules like a solar power bank charger. It converts sunlight into electrical energy and is typically used to charge a 3.7V lithium-ion battery through a charging circuit. These panels are usually made of monocrystalline or polycrystalline silicon, offering good efficiency in converting solar energy. Under ideal sunlight, a typical 5V mini panel can provide a current between 100 mA to 300 mA, corresponding to about 0.5 W to 1.5 W of power. However, real-world performance depends on sunlight intensity, orientation, and weather conditions. A higher-wattage panel (around 2 W–3 W) is preferred for faster charging. The panel usually comes in small sizes such as 70 mm × 70 mm or 107 mm × 61 mm and is lightweight, making it ideal for portable applications. In a solar power bank system, the panel's output is regulated by a charging module that ensures proper charging of the 18650 lithium battery, which then stores energy for later use to charge mobile devices. Overall, a 5V mini solar panel is an eco-friendly and practical solution for low-power renewable energy applications.

4. USB Cabel

A USB cable (Universal Serial Bus cable) is a standard interface used for data transfer and power supply between electronic devices. In a solar power bank charger system, the USB cable plays a vital role in connecting the power bank to mobile phones or other gadgets for charging. It typically consists of two main parts — an inner conductor for data and power transmission and an outer insulating sheath for protection. The cable carries a regulated 5V DC supply from the power bank's output port (usually a Type-A or Type-C connector) to the device being charged.

Most USB cables have four internal wires: two for power (VCC and GND) and two for data (D+ and D–). In solar power banks, the charging function mainly uses the power lines, while data lines are used if communication with a computer or smart device is needed. The most common types of USB connectors are USB-A, Micro-USB, and USB Type-C, with Type-C being the most advanced, capable of faster charging and reversible connection.

In the solar charging circuit, the USB cable connects the output of the 134N3P C-type USB power bank module to the mobile phone, allowing safe and efficient energy transfer. High-quality USB cables are designed to minimize voltage drops, ensuring that the 5V output from the battery reaches the device without significant loss. They are made of durable materials such as copper for better conductivity and are shielded to prevent electromagnetic interference. Overall, the USB cable serves as the essential link between the energy stored in the power bank and the mobile device, enabling convenient and portable solar-powered charging.

Methodology

The methodology for developing the solar power bank charger for mobile involves a systematic approach that includes the selection of components, circuit design, assembly, and testing. The main aim of this methodology is to design a compact and efficient solar-based charging system capable of storing energy and supplying regulated power to charge mobile devices through a USB interface.

1. System Design Overview

The solar power bank system is designed to harness solar energy, convert it into electrical energy through a 5V solar panel, and store it in a 3.7V 18650 lithium-ion battery. The stored energy is then regulated and supplied to a mobile device via the 134N3P C-Type USB Power Bank Module. The major components integrated into the system include the 5V mini solar panel, lithium-ion rechargeable battery, charging and protection module (134N3P), and standard USB cable for output.

2. Working Principle

The operation of the system is based on the photovoltaic effect, where the 5V solar panel converts sunlight into direct current (DC) electricity. This electrical energy is then fed to the 134N3P module, which regulates the charging current and voltage suitable for the 3.7V 18650 lithium-ion battery. Once the battery is fully charged, it acts as an energy reservoir. When a mobile phone is connected via a USB cable, the same module boosts the stored 3.7V DC to a stable 5V DC output required by the mobile phone. This ensures safe and efficient charging even when sunlight is unavailable.

3. Components and Their Integration

The 5V solar panel serves as the primary energy source, providing input power during daylight. The output from the solar panel is connected to the 134N3P C-Type USB Power Bank Module, which contains built-in charging and protection circuits. This module manages the charging process of the 3.7V 18650 3000mAh lithium-ion rechargeable battery, preventing overcharging, over-discharging, and short circuits.

The output of the module is provided through a USB port, allowing connection to mobile devices via a USB cable. The USB cable ensures efficient power transfer with minimal voltage drop. The energy stored in the battery can then be used to charge mobile phones or other 5V gadgets even when sunlight is unavailable.

4. Circuit Assembly

The circuit assembly involves connecting the positive and negative terminals of the solar panel to the input terminals of the 134N3P module. The lithium-ion battery is connected to the battery terminals of the module, ensuring proper polarity. The module’s output terminal (USB port) provides 5V regulated power to the connected mobile device. The setup is assembled neatly on a small board or within an enclosure to form a portable solar power bank unit. The components are tested for proper connectivity and voltage levels using a multimeter.

5. Testing and Performance Evaluation

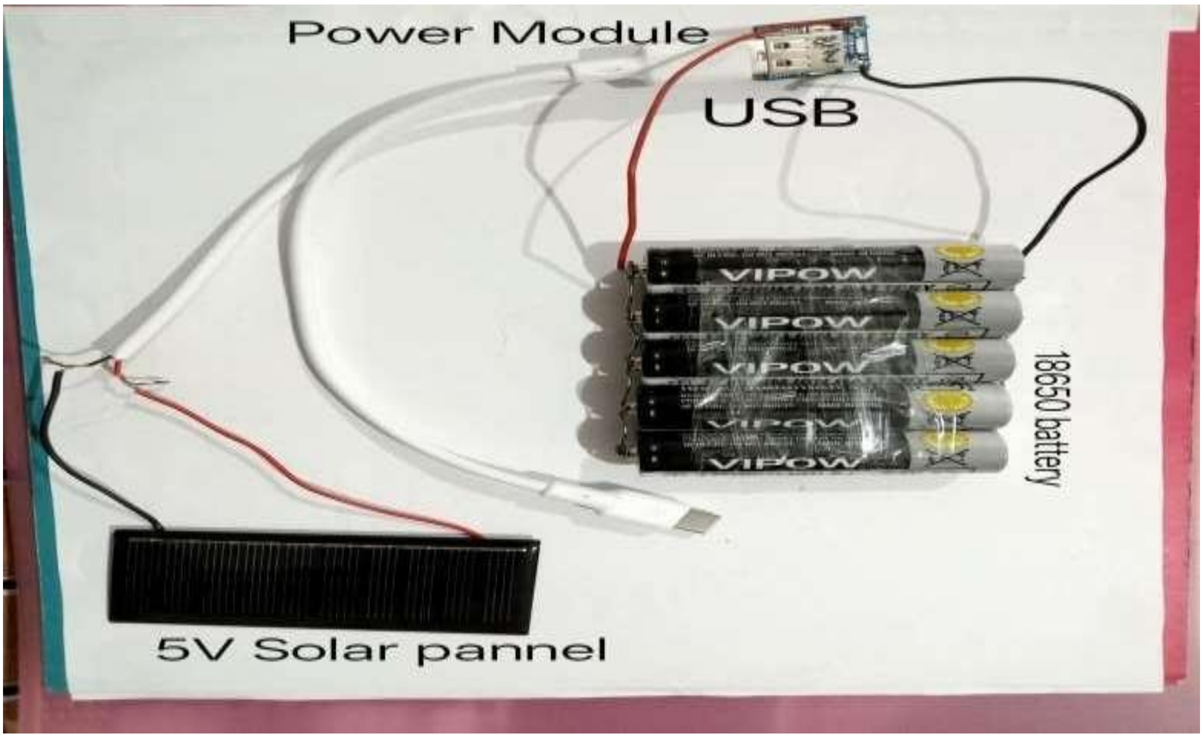
After assembly, the system is tested under direct sunlight to verify its charging capability. The open-circuit voltage of the solar panel and the charging voltage across the battery are measured. The output voltage from the USB port is checked to confirm a stable 5V supply suitable for mobile charging. Charging time, efficiency, and backup duration are evaluated to determine the system's effectiveness. The performance is compared under different sunlight intensities to observe variations in charging speed.

6. Safety and Protection Considerations

Safety is maintained by using the built-in protection features of the 134N3P module, which prevents damage due to over-voltage, over-current, or reverse polarity. The lithium-ion battery is securely housed to avoid exposure to excessive heat or moisture. A blocking diode may also be used to prevent reverse current flow from the battery to the solar panel at night.

7. Expected Output and Applications

The designed system is expected to deliver a 5V DC output at the USB port, sufficient to charge smartphones and small electronic devices. It provides a sustainable, renewable, and portable power source, especially useful in remote areas, outdoor activities, and during power outages.



Conclusion

The development of a **solar-powered power bank integrated with a Battery Management System (BMS)** represents a significant step toward sustainable and portable energy solutions. This project successfully demonstrates how renewable energy can be harnessed effectively to charge modern electronic devices such as smartphones, tablets, and cameras without relying on conventional electricity sources. The use of a **solar panel** allows for energy generation from sunlight, making the system eco-friendly and cost-efficient over time.

The incorporation of a **Battery Management System (BMS)** plays a crucial role in ensuring the safety, reliability, and longevity of the lithium-ion batteries used in the power bank. The BMS continuously monitors critical parameters such as batteryvoltage, current, temperature, and stateof charge, preventing harmful conditions like **overcharging, over-discharging, and short circuits**. This not only safeguards the battery from damage but also enhances its overall performance and lifespan.

Furthermore, the system's design emphasizes **portability, simplicity, and practicality**, making it suitable for remote areas, outdoor activities, and emergency situations where grid power is unavailable. It encourages the use of **clean and renewable energy**, contributing to environmental conservation and reducing carbon footprint.

Through this project, it has been demonstrated that integrating solar energy with modern electronic systems can lead to efficient and sustainable solutions. The knowledge gained from designing and implementing this power bank can be further applied to larger-scale renewable energy systems.

In conclusion, the **solar power bank with BMS** is a reliable, safe, and environmentally responsible device that promotes the efficient use of solar energy. It serves as a promising example of how technology and renewable energy can work hand-in-hand to create a greener future. With further improvements in solar efficiency and battery technology, such systems have the potential to become a mainstream solution for portable power generation and storage.